2D Arrays

LEVEL 6: Special Matrix Checks

Program 58: Check if Square Matrix Write a program that checks if matrix is square (rows = columns).

Program 59: Check if Symmetric Matrix Write a program that checks if matrix equals its transpose.

Program 60: Check if Skew-Symmetric Matrix Write a program that checks if transpose equals negative of matrix.

Program 61: Check if Identity Matrix Write a program that checks if matrix is identity (1s on diagonal, 0s elsewhere).

Program 62: Check if Diagonal Matrix Write a program that checks if all non-diagonal elements are zero.

Program 63: Check if Upper Triangular Matrix Write a program that checks if all elements below diagonal are zero.

Program 64: Check if Lower Triangular Matrix Write a program that checks if all elements above diagonal are zero.

Program 65: Check if Sparse Matrix Write a program that checks if matrix is sparse (more than half elements are zero).

Program 66: Check if Dense Matrix Write a program that checks if matrix is dense (few zero elements).

Program 67: Check if Orthogonal Matrix Write a program that checks if matrix multiplied by its transpose gives identity matrix.

Program 68: Check if Magic Square Write a program that checks if all rows, columns, and diagonals have same sum.

Program 69: Calculate Magic Constant Write a program that calculates magic constant for magic square: $n(n^2+1)/2$.

Program 70: Check if Toeplitz Matrix Write a program that checks if matrix is Toeplitz (diagonal elements are same).

Program 71: Check if Hankel Matrix Write a program that checks if matrix is Hankel (anti-diagonals are constant).

Program 72: Density of Matrix (Ratio of Non-Zero Elements) Write a program that calculates density: (count of non-zero / total elements) × 100.

LEVEL 7: Matrix Patterns & Traversals

Program 73: Print Matrix in Row-Major Order Write a program that prints matrix elements row by row in single line.

Program 74: Print Matrix in Column-Major Order Write a program that prints matrix elements column by column.

Program 75: Print Matrix in Spiral Order (Clockwise) Write a program that prints elements in spiral from outside to inside clockwise. Example: [[1,2,3],[4,5,6],[7,8,9]] → 1,2,3,6,9,8,7,4,5.

Program 76: Print Matrix in Spiral Order (Anti-Clockwise) Write a program that prints spiral in counter-clockwise direction.

Program 77: Print Matrix in Zig-Zag (Snake) Pattern Write a program that prints: row 0 left-to-right, row 1 right-to-left, alternating.

Program 78: Print Matrix in Wave Pattern (Column-wise) Write a program that prints: col 0 top-to-bottom, col 1 bottom-to-top, alternating.

Program 79: Print Matrix Diagonally (Top-Right to Bottom-Left) Write a program that prints all diagonals parallel to secondary diagonal.

Program 80: Print Matrix Diagonally (Top-Left to Bottom-Right) Write a program that prints all diagonals parallel to main diagonal.

Program 81: Print Boundary Layer by Layer Write a program that prints outer boundary, then next inner boundary, and so on.

Program 82: Fill Matrix in Spiral Order Write a program that fills matrix with numbers 1 to N^2 in spiral pattern.

Program 83: Generate Identity Matrix Write a program that creates and displays N×N identity matrix.

Program 84: Generate Zero Matrix Write a program that creates M×N matrix filled with zeros.

Program 85: Generate Matrix with Random Numbers Write a program that fills matrix with random numbers in given range.

Program 86: Generate Diagonal Matrix Write a program that creates diagonal matrix from given diagonal elements.

Program 87: Print All Saddle Points Write a program that finds saddle points (minimum in row AND maximum in column).

LEVEL 8: Searching in Matrix

Program 88: Linear Search in Matrix Write a program that searches element using linear search row-by-row.

Program 89: Search in Row-wise Sorted Matrix Write a program that searches in matrix where each row is sorted independently.

Program 90: Search in Column-wise Sorted Matrix Write a program that searches in matrix where each column is sorted.

Program 91: Search in Row & Column Sorted Matrix (Staircase Search) Write a program that searches in matrix sorted both row-wise and column-wise. Start from top-right or bottom-left.

Program 92: Binary Search in Each Row Write a program that applies binary search on each row to find element.

Program 93: Find Row with Given Element Write a program that finds which row contains given element (return first occurrence).

Program 94: Find Column with Given Element Write a program that finds which column contains given element.

Program 95: Count Occurrences in Entire Matrix Write a program that counts total occurrences of element in matrix.

Program 96: Find Element with Maximum Occurrences Write a program that finds which element appears most frequently.

Program 97: Search in Fully Sorted Matrix Write a program that searches in matrix where elements increase row-wise and column-wise (treat as 1D sorted array).

Program 98: Find Kth Smallest Element in Matrix Write a program that finds Kth smallest element in sorted matrix.

Program 99: Find Median in Row-wise Sorted Matrix Write a program that finds median of all elements in row-wise sorted matrix.

LEVEL 9: Advanced Matrix Problems

Program 100: Set Matrix Zeros Write a program that sets entire row and column to 0 if any element is 0. Example: If matrix[i][j]=0, set matrix[i][*]=0 and matrix[*][j]=0.

Program 101: Set Matrix Zeros (In-Place, O(1) Space) Write a program that performs above operation without using extra matrix.

Program 102: Row with Maximum 1s (Binary Matrix) Write a program that finds row with maximum number of 1s in binary matrix where rows are sorted.

Program 103: Count 1s in Binary Matrix Write a program that counts total number of 1s efficiently in row-sorted binary matrix.

Program 104: Find Maximum 1s Rectangle Write a program that finds largest rectangle of all 1s in binary matrix.

Program 105: Island Count (Connected 1s) Write a program that counts number of islands (groups of connected 1s) in binary matrix.

Program 106: Largest Square of All 1s Write a program that finds size of largest square sub-matrix with all 1s.

Program 107: Rotate Matrix 90° In-Place Write a program that rotates square matrix 90° clockwise without extra space.

Program 108: Print Matrix in Anti-Spiral Write a program that prints from center outward in spiral.

Program 109: Hourglass Sum in Matrix Write a program that finds maximum hourglass sum (pattern: 3 top + 1 middle + 3 bottom).

Program 110: Maximum Path Sum (Top-Left to Bottom-Right) Write a program that finds maximum sum path from top-left to bottom-right (can move only right or down).

Program 111: Minimum Path Sum Write a program that finds minimum sum path from top-left to bottom-right.

Program 112: Count Paths (Top-Left to Bottom-Right) Write a program that counts total number of unique paths from start to end.

Program 113: Unique Paths with Obstacles Write a program that counts paths avoiding obstacles (marked as -1).

Program 114: Determinant of Matrix (2×2 and 3×3) Write a program that calculates determinant for 2×2 and 3×3 matrices.

Program 115: Check if Matrix is Singular Write a program that checks if determinant is zero (singular matrix).

Program 116: Trace of Matrix Write a program that calculates trace (sum of main diagonal).

Program 117: Norm of Matrix (Frobenius Norm) Write a program that calculates Frobenius norm: √(sum of squares of all elements).